

CHAPTER 2

Steel Fundamentals

STEELS are among the most widely used metals and have a wide variety of applications. Traditionally, steels are used to build cars, appliances, machines, and structures. More advanced specialty steels are used for corrosion resistance (stainless steel), high-temperature applications (super alloys), and lightweighting (high-strength steel, HSS). New grades of advanced high-strength steels (AHSS) have been developed or are still under development for use in applications where strength, safety, recyclability, and cost are essential. Steel has the widest range of compositions, microstructures, and properties of most metal alloys and, therefore, is suited to a wide range of applications.

2.1 Compositions and Metallurgical Phases

Steel is a versatile metal alloy that is basically iron and carbon with varying small amounts of other alloying elements such as manganese, chromium, nickel, and molybdenum. Some of the elements are residual while others are added in order to impart useful properties such as strength, ductility, hardness, toughness, wear resistance, machinability, and weldability. In carbon steel, up to 2% C interacts with iron to produce two principal phases: a single-phase solid solution of body-centered cubic (bcc) iron that is soft and ductile, called *ferrite* (α), and another phase composed of a chemical compound that is hard and strong, called *cementite* or iron carbide (Fe_3C). Cementite forms in the presence of high carbon concentrations. A two-phase mixture called *pearlite* forms at lower carbon content and is made up of alternating layers of soft ferrite and hard cementite. *Martensite* is a very hard phase that results from the very rapid quenching of *austenite* (γ), which is the high-temperature phase of the Fe-C solid solution. *Bainite* is a strong phase that has the same composition as that of pearlite but forms when austenite is cooled more quickly than necessary to form pearlite, but not quickly enough to form martensite.

The basic guide to understanding the behavior of carbon steel is the Fe-C phase diagram. This diagram is fundamental to understanding the relationship between heat treatment, microstructure, and mechanical properties of iron-carbon alloys. Figure 2.1 shows part of the phase diagram for compositions of carbon steels with <2% C and temperatures below 1000 °C (1830 °F). This range defines the wrought steels that are significant for most engineering applications. The diagram shows how temperature and carbon content combine to give a variety of metallurgical structures. Carbon atoms dissolve in iron as interstitial impurities to form the solid solution known as ferrite. When the solubility limit of carbon in iron is exceeded, the chemical compound cementite, or iron carbide (Fe_3C), forms. Of special interest in the Fe-C phase diagram is the eutectoid composition that occurs at 0.76% C. At this composition, the high-temperature austenite will undergo the eutectoid reaction (austenite $\rightarrow$ ferrite + cementite) at 720 °C (1330 °F). The eutectoid steel is composed of 100% pearlite whose microstructure consists of alternate layers of ferrite and cementite.

At room temperature, the stable phases of steel are ferrite and cementite. Ferrite is a common constituent in steels and has a less densely packed bcc structure. Cementite is a very hard and brittle chemical compound that enhances the strength of steels. All steel alloys, up to 2%, must cool through the γ phase, austenite. Austenite is a high-temperature phase and has a close-packed face-centered cubic (fcc) structure. Heating carbon steel into the austenite region is called austenitization. Figure 2.1 shows that the austenitization temperature decreases with carbon content up to the eutectic composition of 0.76% C, then increases for higher carbon content up to 2% C. Austenite is nonmagnetic and can be retained at room temperature by the addition of appropriate alloying elements such as manganese and nickel.

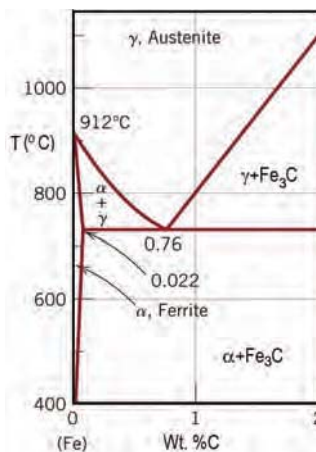


Fig. 2.1 Portion of the Fe- Fe_3C phase diagram for carbon steel

Ferrite is iron containing a very small amount of carbon. The maximum solubility of carbon in iron is 0.022% at 725 °C. The iron-carbon phase diagram describes what happens when a certain grade of carbon steel is slowly heated or cooled. For instance, eutectoid steel transforms completely to austenite on heating above the eutectoid temperature of 725 °C and transforms completely to pearlite on cooling below this temperature. The carbon content provides many different steel compositions and microstructures and, therefore, is the major factor in determining the properties that can be developed in steel. For example, strength increases with carbon content up to the eutectoid composition but then starts to drop as a grain-boundary network of brittle cementite is formed.

Most of the carbon in steel is present in the microstructure as iron carbide. Hence, the wide range of properties in steel that has not been heat treated is determined primarily by the relative amounts of iron and iron carbide in the microstructure. Increasing the fraction of iron carbide in steel, while holding microstructural features constant, will produce harder and stronger steel. Although carbon is the element that transforms iron into steel, other elements are also added to produce a variety of desirable properties. The effect of the alloying elements on the Fe-C phase diagram is to shift its phase boundaries and the shapes of the phase areas. Figure 2.2 shows the dependence of the eutectoid temperature on alloy concentration for several alloying elements in steel. The figure shows that the eutectoid reaction, which occurs at 725 °C for unalloyed steel, is greatly influenced by the type and concentration of the alloying elements.

2.2 Microstructure and Heat Treatment

The three major variables in steel processing are alloying, work hardening, and thermal processing. Alloying refers to the ability of iron to form

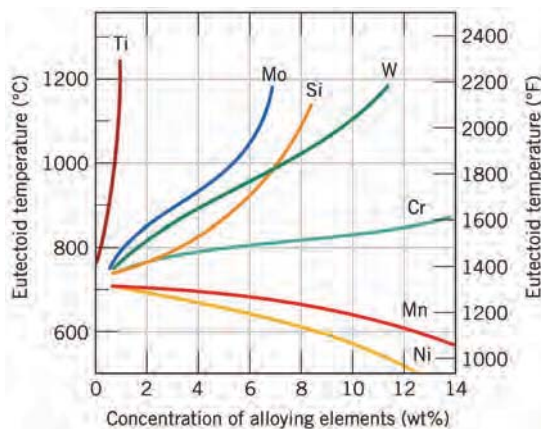


Fig. 2.2 Effect of alloy concentration on the eutectoid temperature in steel.
Source: Ref 2.1

solid solutions and compounds with other metals. Work hardening refers to the tendency of steel to become stronger when subjected to plastic deformation. Thermal processing refers to the ability of steel to change its mechanical properties by heat treating while in the solid state. Although chemistry and plastic deformation are important to steel manufacture, heating and cooling are essential for phase transformation and the development of microstructures. Steel properties and performance depend on its microstructure, which is a function of processing.

Heat treatment involves moving atoms around within the crystalline structure of metal by means of solid-state diffusion. The mechanism of diffusion relies on the presence of vacancies in the crystal lattice. The basis of heat treatment relies on the fact that the solubility of carbon in austenite (fcc) is much higher than in ferrite (bcc) because the size of interstices that can accommodate carbon atoms are larger in the fcc structure. At room temperature, all carbon steels have two phases (ferrite and iron carbide). Austenitizing steel produces a single phase (α phase) and this is used as the starting point to control microstructure by heat treating procedures. For carbon steels, cooling transforms the single phase to two phases of widely different compositions. For alloy steels that have more sophisticated chemical compositions, more complex phases are produced.

Iron carbide particles embedded in the ferrite grains form a microstructure that can be observed under a microscope. Figure 2.3 shows the microstructure of eutectoid steel consisting of alternating lamellae of two phases (α and Fe_3C).

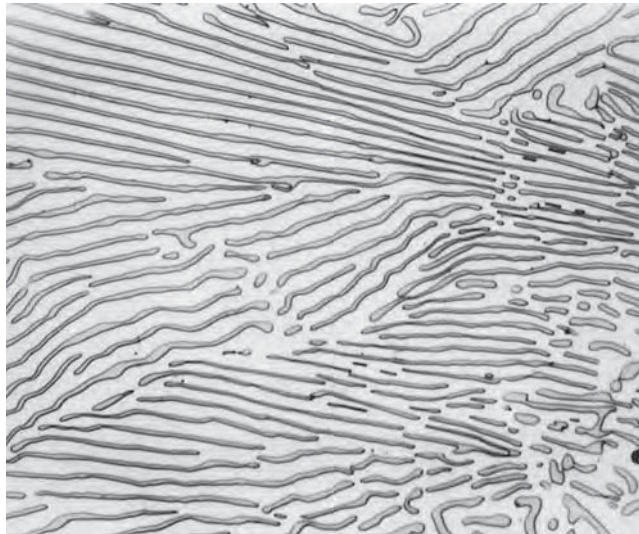


Fig. 2.3 Microstructure of pearlite in eutectoid steel consisting of alternating layers of α -ferrite (light phase) and iron carbide (dark layers). Source: Ref 2.1

Microstructural features include not only the relative amounts of the constituents but also their nature, size, shape, and distribution. For a given carbon content, these features are brought about by heat treatment and can result in significant changes in the properties of carbon steel. The role of heat treatment is to produce a final microstructure in steel that will result in desired properties for the intended application. Microstructure development can be achieved by heat treating cycles aimed at controlling the precipitation of iron carbide from a solid solution of austenite.

Austenite is the high-temperature phase of steel that decomposes on cooling to form the final microstructure of the steel. For a specific steel composition, this microstructure depends on the temperature at which the precipitation occurs. Under equilibrium conditions, austenite transforms to pearlite at a temperature just below the critical temperature of 725 °C (1340 °F).

If nonequilibrium conditions exist (rapid or slow cooling), the transformation microstructure varies markedly with the temperature of the transformation. The transformation of austenite is studied by observing the constant temperature transformation behavior at a series of temperatures below the critical temperature of 725 °C. The results of these observations are presented as a chart where the times at which transformation begins and at which it goes to completion, at a given temperature, are plotted. Such a diagram is referred to as a time-temperature-transformation (TTT) diagram and is shown in Fig. 2.4 for eutectoid steel.

Figure 2.4 also shows how the TTT diagram is generated from individual percent transformation versus logarithm of time plots obtained after the rapid cooling of a sample to a specified constant temperature. The shape of the curve results from a diffusion-limited transformation at low temperatures and an energy-limited transformation at high temperatures. At high temperature, diffusion is fast and coarse pearlite forms. At low temperatures, diffusion is slow and fine pearlite forms, but only near the “knee” of the TTT diagram. Below the “knee,” transformation produces a feathery microstructure called bainite.

Figure 2.5 shows a superimposed isothermal heat treatment curve (ABCD) where austenite begins its transformation at $T > 725$ °C. It is then rapidly cooled to $T = 625$ °C (1160 °F) and then held at constant temperature of 625 °C until decomposition takes place. This isothermal treatment produces the fine pearlite structure shown in Fig. 2.5. The temperature around 550 °C (1020 °F) gives the shortest time (incubation period) for starting the austenite-to-pearlite transformation and is referred to as the “knee” of the curve, N , as shown in Fig. 2.6. This location also defines the critical cooling rate of the steel.

A more complete isothermal transformation diagram for austenite-to-pearlite (A-P) and austenite-to-bainite (A-B) transformation regions is shown in Fig. 2.6. While pearlite forms above the “knee” of the curve, N , bainite, which consists of a very fine combination of ferrite and carbide,

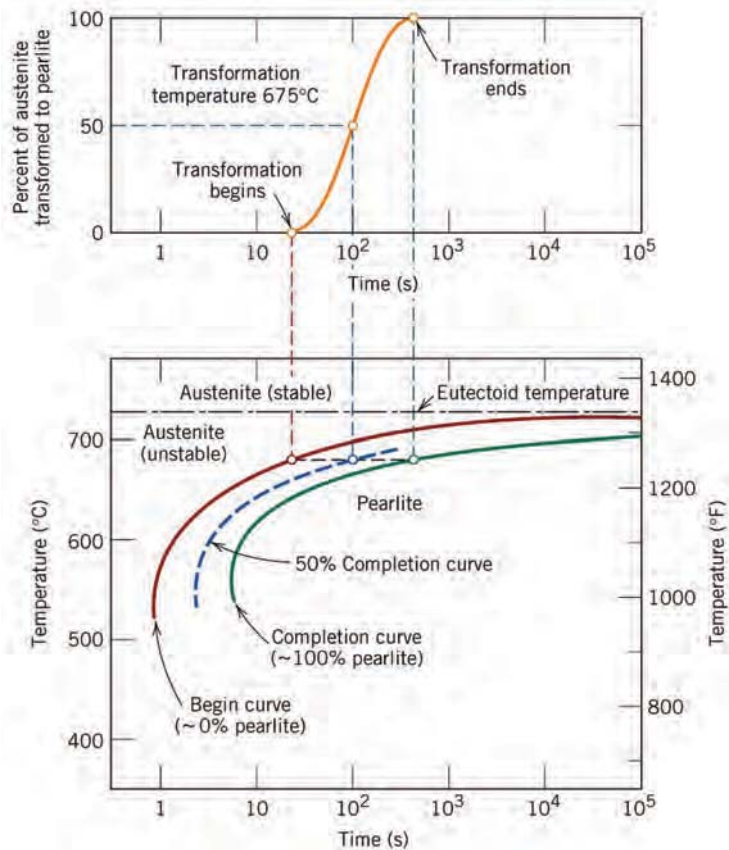


Fig. 2.4 Generation of time-temperature-transformation (TTT) diagram from percent of transformation vs. log time plots. Source: Ref 2.1

forms below it. Pearlite forms in the region from the eutectoid temperature 725 °C (1340 °F) down to approximately 550 °C (1020 °F). Below that temperature, pearlite does not form and, instead, bainite forms. In general, the transformed microstructure becomes more fine grained as the transformation temperature is decreased. Increasing the carbon and alloy content decreases the transformation rate and affects the shape of the TTT diagram by shifting it to the right. This means that the transformation at all temperature levels starts later and takes more time to completion.

Most heat treating processes for steel involve continuous cooling to room temperature rather than isothermal heat treatment. For continuous cooling, the time required to start the transformation is delayed. Therefore, the isothermal transformation curves will be slightly shifted to longer times and lower temperatures. Such curves are referred to as continuous cooling transformation (CCT) diagrams. Figure 2.7 shows the superposition of the CCT diagram on the isothermal cooling (TTT) diagram.

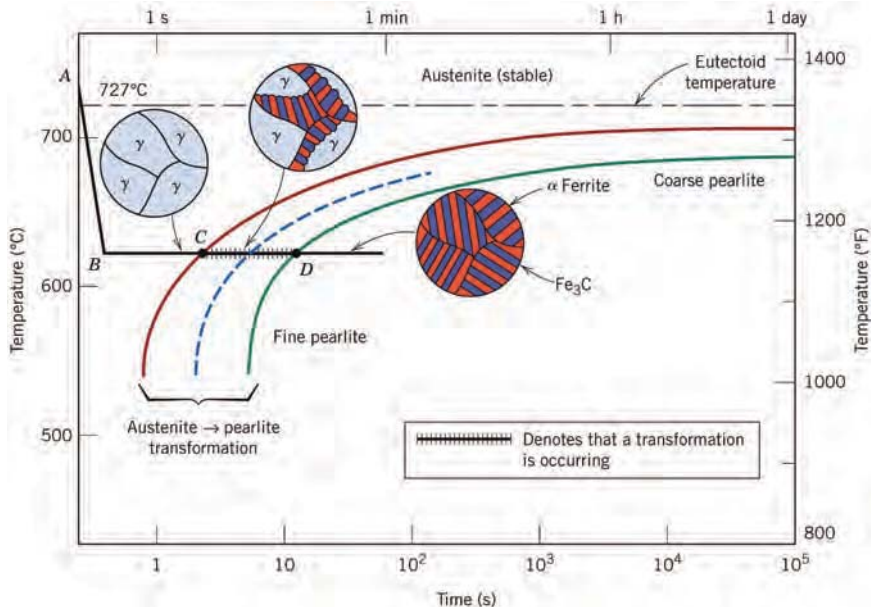


Fig. 2.5 Isothermal transformation of austenite to pearlite in eutectoid carbon steel. Source: Ref 2.1

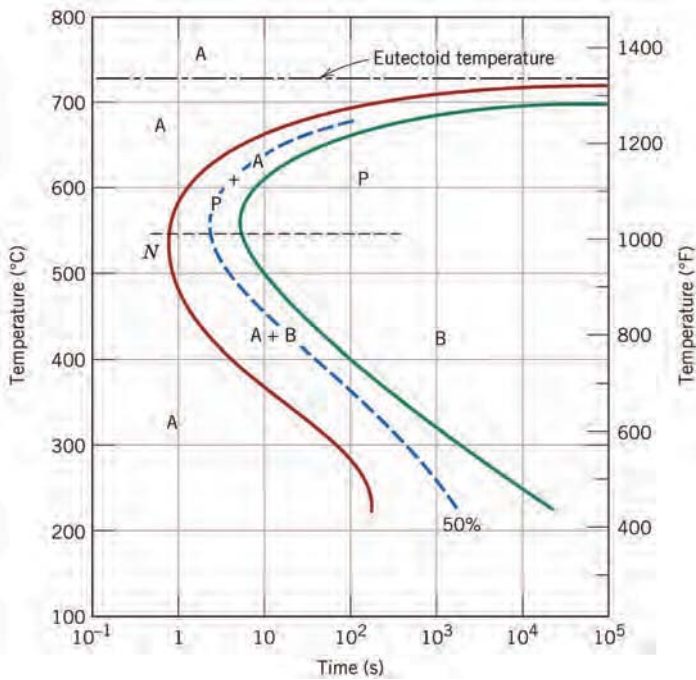


Fig. 2.6 Isothermal transformation of eutectoid steel from austenite to pearlite (A-P) and austenite to bainite (A-B). Source: Ref 2.1

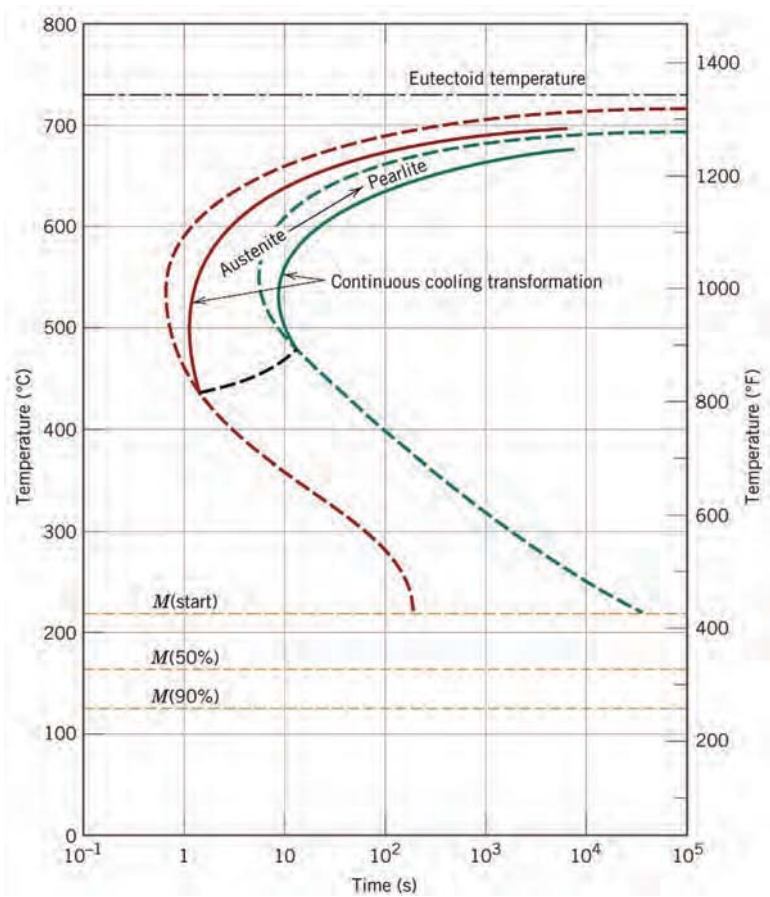


Fig. 2.7 Relationship between isothermal cooling diagram (TTT) and continuous cooling diagram (CCT) for eutectoid steel. Source: Ref 2.1

A schematic representation of the temperature histories for an isothermal cooling diagram (TTT) and a CCT diagram is shown in Fig. 2.8. The temperature histories for the TTT diagram in Fig. 2.8(a) show two quench and hold isothermal treatments, *A* and *B*, to room temperature, *RT*. At *A*, 50% of the transformation from austenite to bainite occurred, while at *B* the transformation is 100% complete. For the CCT diagram in Fig. 2.8(b), three continuous cooling treatments for three different cooling rates are shown. The middle cooling rate is known as the critical cooling rate (CCR) and is defined as the minimum rate that avoids the start of any diffusion-controlled transformation.

Figure 2.9 shows cooling curves for moderately fast (10 min) and slow (15 h) cooling times superimposed on a CCT diagram of eutectoid steel. The fast cooling rate produces a microstructure of fine pearlite and the slow cooling rate produces one with coarse pearlite. The explanation for this is based on the fact that low nucleation rates and high diffusion rates

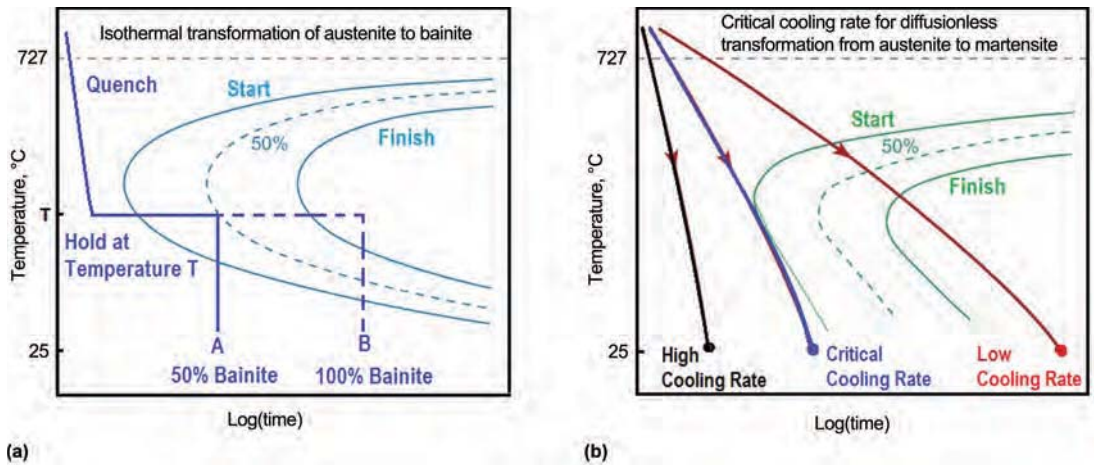


Fig. 2.8 Schematics showing the temperature histories of (a) an isothermal cooling diagram (TTT), and (b) a continuous transformation diagram (CCT). Source: Adapted from Ref 2.2

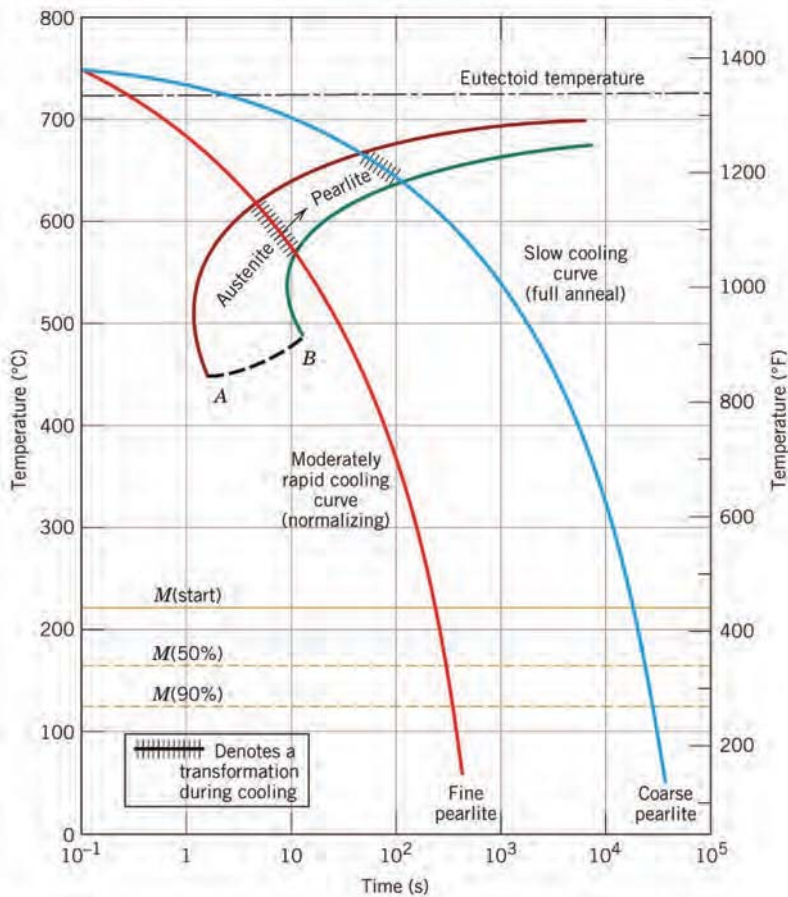


Fig. 2.9 Microstructures of eutectoid steel resulting from fast and slow continuous cooling rates. Source: Ref 2.1

lead to coarse structures while high nucleation rates and low diffusion rates lead to fine structures.

The microstructures of coarse and fine pearlite for eutectoid steel are shown in Fig. 2.10. Coarse pearlite is characterized by thick and widely spaced lamellae of ferrite and cementite while the lamellae of fine pearlite are thin and closely packed. Fine pearlite is harder and less ductile than coarse pearlite.

In many situations, the objective of heat treatment is to cool the sample rapidly enough so that transformation does not occur at all. Close inspection of the CCT diagram in Fig. 2.7 indicates that no information is available below approximately 250 °C (480 °F). Three horizontal lines are added below that temperature to show the transformation of quenched austenite to martensite. Martensite is the microstructure obtained by the rapid quenching of austenite to bypass the “knee” of the CCT diagram. Martensite is important because it is the hardest structure obtained from the transformation of austenite. Figure 2.11 shows a micrograph of martensitic microstructure. As formed, martensite is of little use and must be softened for metalworking. Softening is accomplished by reheating or tempering martensite at temperatures below the eutectoid temperature of 725 °C (1340 °F). Properties of tempered martensite depend on the tempering temperature and the hold time at that temperature.

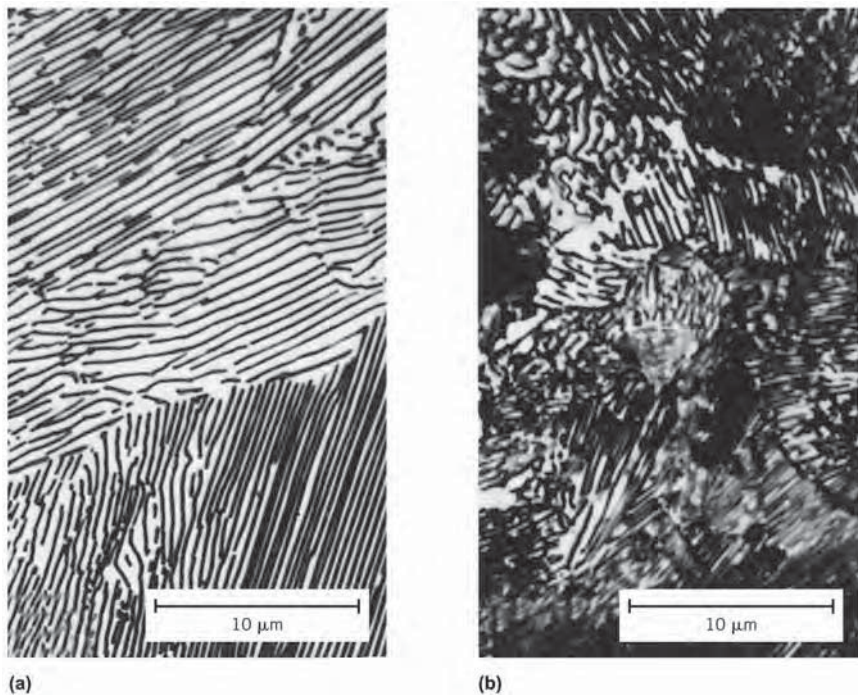


Fig. 2.10 Micrographs of (a) coarse pearlite and (b) fine pearlite of eutectoid steel. Source: Ref 2.1



Fig. 2.11 Martensite microstructure (needle-shaped grains) and retained austenite (white matrix). Source: Ref 2.1

Martensitic transformation involves the reorientation of carbon and iron atoms from the fcc solid solution of austenite to a distorted body-centered tetragonal (bct) solid solution of martensite. The distortion is caused by trapped atoms of carbon and the distorted lattice is resistant to dislocation glide and hence increases its strength and hardness. Transformation to martensite occurs at temperatures below 250 °C, and the percentage of transformation depends only on the temperature to which it is cooled. Transformation to martensite occurs by a diffusionless process, and it differs from transformation to pearlite or bainite in that it is not time dependent, but occurs instantly during cooling. Figure 2.12 shows the CCR for eutectoid steel superimposed on a CCT diagram. It also demonstrates the dependence of the final microstructure on the cooling rate. Cooling rates greater than the critical cooling rate will produce martensite only. Lower cooling rates than the critical will produce both martensite and pearlite, and for the lowest cooling rates only pearlite forms.

The transformation from austenite to martensite produces a brittle microstructure with low fracture toughness and diminished ability for deformation processing of the material. Ductility is regained by a reheating process called tempering. On tempering, the excess carbon in the deformed bct lattice combines with iron atoms to form Fe_3C precipitates.

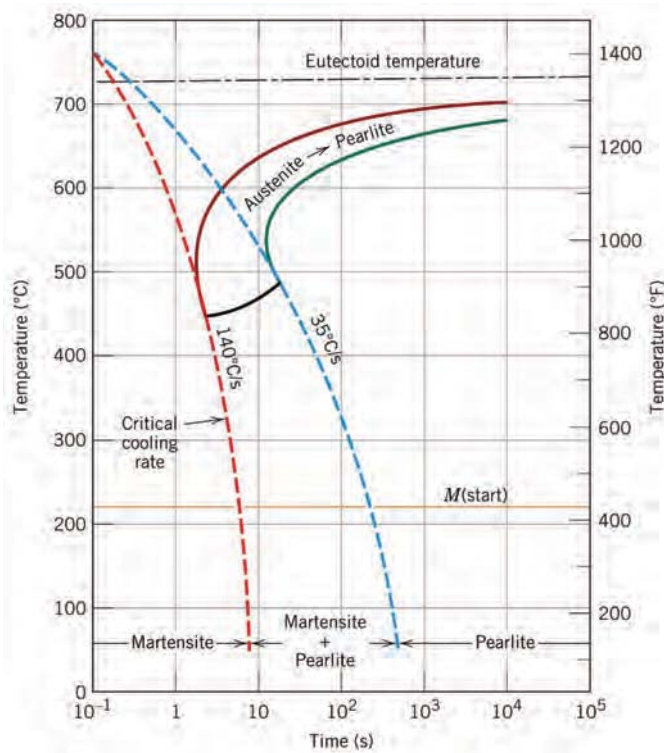


Fig. 2.12 Effect of cooling rate on development of final microstructure.
Source: Ref 2.1

The strengthening mechanism in tempered martensite shifts from resistance to dislocation glide in a distorted lattice to resistance to dislocation glide by precipitates. The resulting yield stress evolves with tempering time because the process of nucleation and growth of the Fe_3C precipitates is time dependent.

Figure 2.13 shows (a) quenching of austenite to martensite on a CCT diagram and (b) evolution of yield stress during tempering of quenched carbon steel. Tempering reduces the yield strength to levels between quenched martensite (2500 MPa, or 363 ksi) and normalized ferrite-pearlite microstructure (350 MPa, or 51 ksi). Manipulating properties of steel by using quench and temper treatment is a widespread technique for controlling the toughness of steel.

A block diagram that summarizes the effect of cooling rates on the microstructure of steel is shown in Fig. 2.14.

A summary of the microstructures and mechanical attributes of carbon steel is listed in Table 2.1.

A major concern in heat treatment is that a low enough cooling rate cannot be practically achieved to avoid the “knee” of the CCT diagram; hence, 100% martensite cannot be produced. This problem can be solved

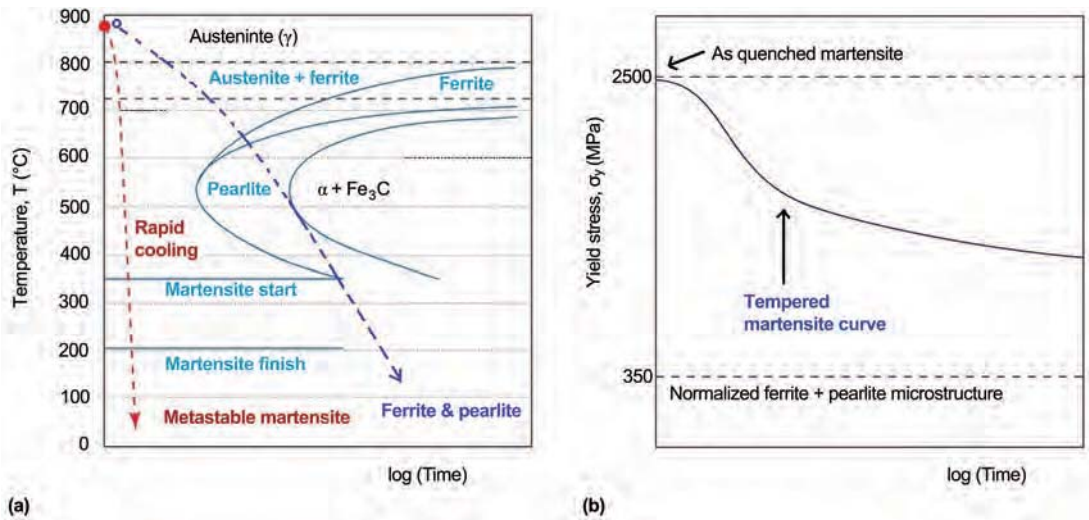


Fig. 2.13 (a) Martensite formation from austenite. (b) Evolution of yield stress during tempering of martensite. Source: Adapted from Ref 2.2

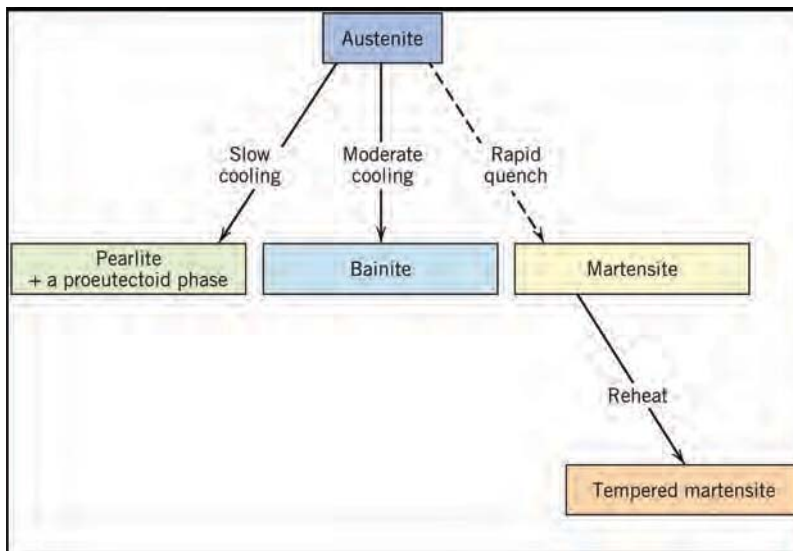


Fig. 2.14 Block diagram of the effect of cooling rate on development of final microstructure. Source: Ref 2.1

Table 2.1 Microstructure and attributes of steel

Microconstituent	Phases present	Mechanical attributes
Pearlite	α ferrite + Fe_3C	Soft and ductile
Bainite	α ferrite + Fe_3C	Stronger and less ductile than pearlite
Martensite	Single-phase bct structure	Stronger and less ductile than bainite
Tempered martensite	α ferrite + fine Fe_3C	Strong but more ductile than martensite

bct, body-centered tetragonal. Source: Compiled from Ref 2.1

by shifting the CCT diagram to the right to achieve longer times for its “knee.” The shift can be accomplished by increasing the carbon content and adding small amounts of alloying elements. Figure 2.15 shows a shifted CCT diagram for high-strength, low-alloy (HSLA) steel AISI/SAE 4340 obtained by increasing carbon content to 0.40% and adding alloying elements nickel, chromium, and molybdenum. Alloying elements affect diffusion and limit austenite transformation at high temperature to the extent that a double “knee” is formed. Figure 2.15 shows that the “knee” of the CCT diagram shifted from a low cooling time of approximately 1 second for low carbon steel (Fig. 2.9) to a high cooling time of approximately 10 seconds for HSLA steel (Fig. 2.15).

The properties of steel can be controlled by heat treating procedures through the control of microstructure. Heat treating involves heating the steel to a temperature where austenite is formed and then cooling it at an appropriate rate to transform it to the desired microstructure. Heat treat-

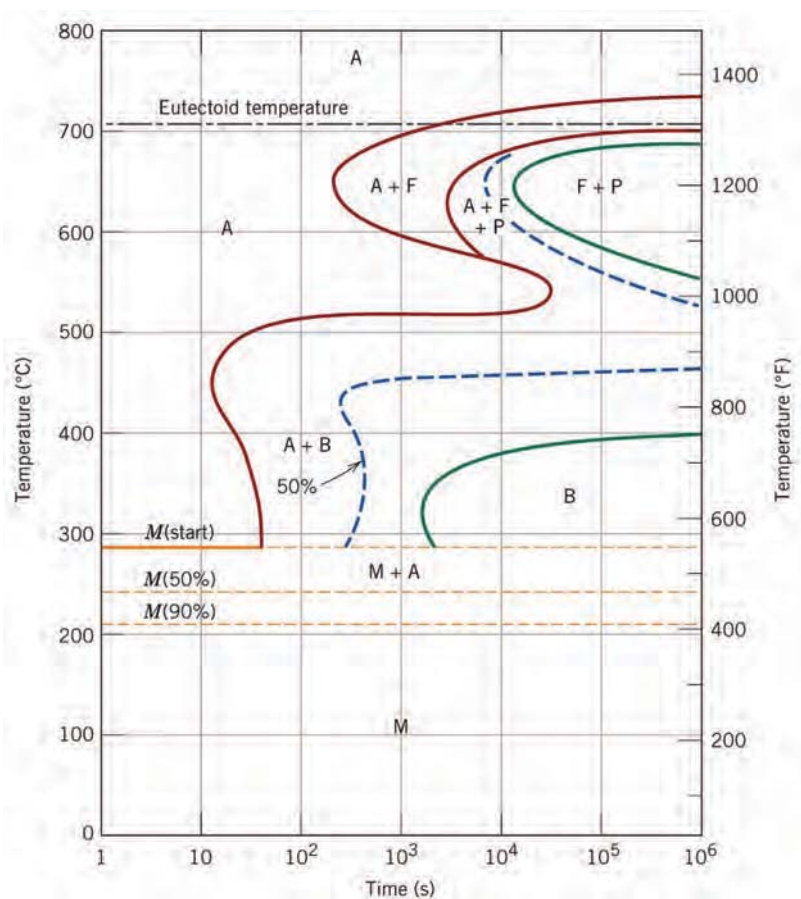


Fig. 2.15 Continuous cooling transformation (CCT) diagram for high-strength, low-alloy steel AISI/SAE 4340. Source: Ref 2.1

ment usually includes reheating and cooling to relieve residual stresses and to modify the microstructure to control the strength and ductility of the steel.

Conventional heat treating procedures include annealing, normalizing, quenching and tempering, stress relieving, martempering, and austempering. Figure 2.16 shows time-temperature diagrams of the major heat treating procedures and products produced by decomposition of austenite. The heat treating diagrams are superimposed on a typical isothermal transformation diagram to illustrate graphically the heating and cooling operations involved in each heat treating process. Each heat treating procedure produces a microstructure that determines the final mechanical properties of the steel. The first step in the heat treating procedures that are based on the decomposition of austenite requires that the steel be heated to a temperature within the austenitizing range of 790 to 915 °C (1455 to 1680 °F). Following is a summary of the four major transformation cycles:

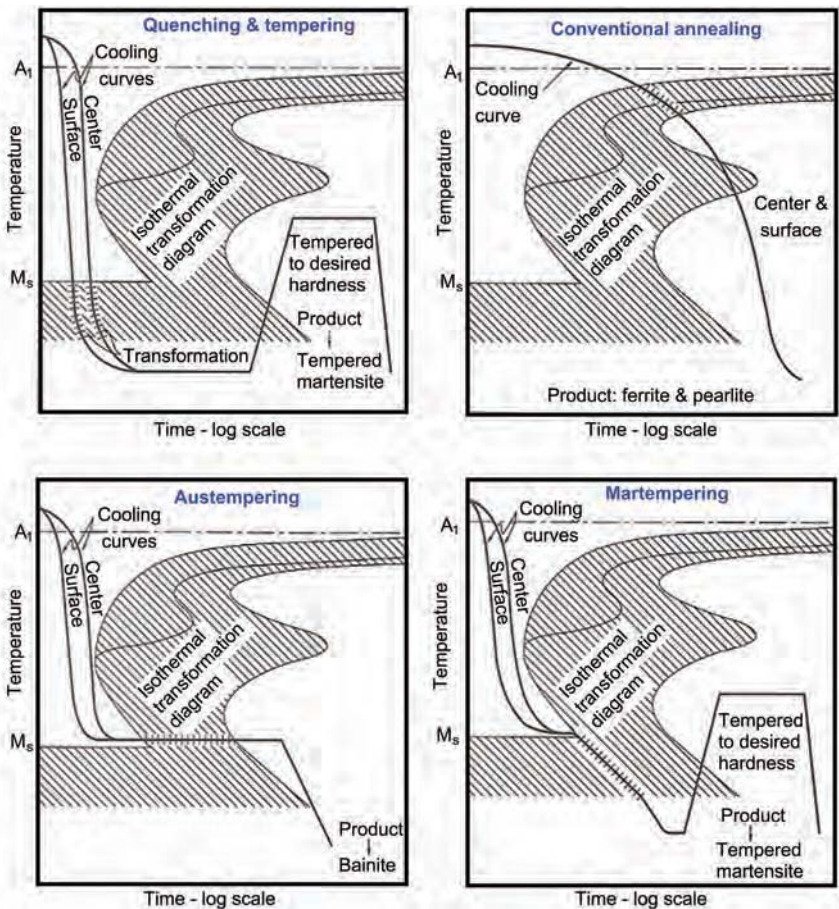


Fig. 2.16 Major heat treating procedures superimposed on TTT diagram.
Source: Ref 2.3

1. *Conventional Quenching and Tempering:* Austenitizing steel disperses the carbon into solution. Quenching to room temperature traps the carbon in a supersaturated solution. The phase transformation of iron from fcc to bcc results in a metastable phase called martensite. The extra carbon in the bcc lattice causes it to distort into the bct lattice of martensite. Martensite has very high yield strength but very low ductility. It is then tempered to desired hardness by heating at temperatures in the range of 210 to 650 °C (410 to 1200 °F) to transform martensite to ferrite and cementite. The microstructure of tempered martensite is different from that of annealed ferrite and cementite.
2. *Conventional Annealing:* Austenitized steel is cooled slowly to form ferrite and pearlite.
3. *Austempering:* Austenitized steel is quenched in a bath maintained at a constant temperature range of 260 to 400 °C (500 to 750 °F). The steel is allowed to transform isothermally to bainite in this bath, then cooled to room temperature.
4. *Martempering:* Austenitized steel is quenched below the “knee” of the TTT diagram. It is then slow cooled to form martensite and tempered to desired hardness by transforming martensite to ferrite and cementite.

2.3 Structure-Property Relationships

The mechanical properties of metals that are of most interest to engineers are strength, ductility, and toughness. Strength determines the stress that can be applied to a structure without resulting in plastic deformation. For most applications the yield strength is important to prevent the onset of plastic flow. For other applications the tensile strength defines the ability for a structure to carry or resist mechanical forces or stresses. Ductility measures the ability of a metal to be formed and is defined as the amount of plastic flow that can occur beyond the yield and before fracture. It is measured in terms of percent elongation or reduction in area. Toughness is related to the resistance of the metal to fracture under conditions of stress concentration resulting from cracks, notches, or defects. It is measured in terms of the energy required to fracture a standard test specimen.

Strength, ductility, and toughness are related because all three are governed by the same deformation mechanism of dislocation glide. In general, as strength increases, ductility and toughness decrease. These properties are microstructure-sensitive because they are influenced by the amount and distribution of constituents and the presence of imperfections such as solutes, inclusions, precipitates, dislocations, twins, grain boundaries, and cracks in the metal.

Mechanical and performance properties of steels are determined by three main factors:

- Composition of constituents
- Microstructures resulting from heat treatment

- Loading mode (tension, compression, and shear)

Engineering components can have complex shapes and can be loaded by a variety of forces. The stresses in small elements within the components can be a combination of three basic modes of loading: tension, compression, and shear. Loading effects, other than tension, are not considered in this chapter. Accordingly, the structure-property relationship in steel is based on the effects of composition and microstructure. Variations in mechanical properties due to steel composition are shown in Fig. 2.17. Increasing the carbon content increases the yield and tensile strengths as well as the hardness of plain carbon steel. Ductility, as measured by elongation or reduction in area, and toughness decrease with increase in carbon content of plain carbon steels.

The microstructure of steel has a significant effect on its strength. For steel with a certain composition the microstructure can be altered by varying the heat treating process. Table 2.2 shows how different yield strengths can be achieved by obtaining different microstructures for a specific steel composition of 0.15 wt% C and 1.0 wt% Mn.

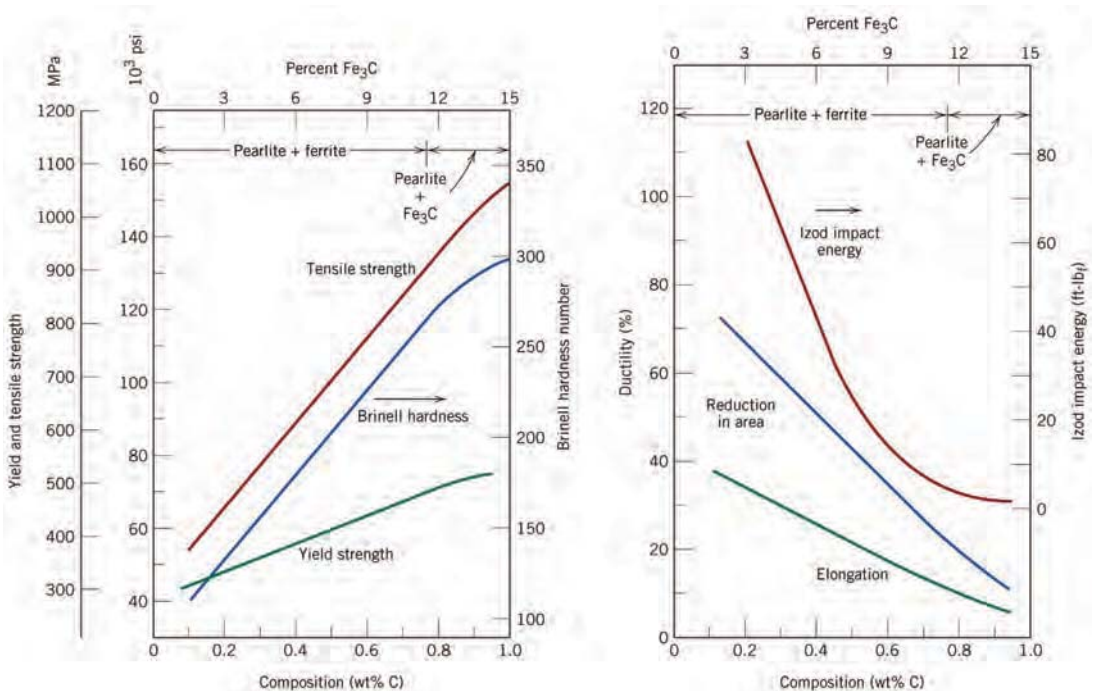


Fig. 2.17 Effect of carbon content on the mechanical properties of plain carbon steels. Source: Ref 2.1

Table 2.2 Effect of microstructure on yield strength of steel

Heat treating process	Microstructure	Yield strength, MPa (ksi)
Water quench	Martensite	450 (65)
Oil quench	Bainite	300 (44)
Air cool	Ferrite + pearlite	150 (22)

Figure 2.18 shows the effect of microstructure on the hardness and ductility of different plain carbon steel compositions. Alloys containing pearlitic microstructures have greater strength and hardness than those containing ductile spheroidite.

Figure 2.19 shows the effect of carbon content on the tensile strength and hardness of carbon steels. High-carbon steels can also be heat treated to form martensite, making the steel harder and stronger.

Different thermal processing schemes produce different room-temperature microstructures. Slow cooling leads to soft ferrite microstructure while quenching produces hard martensite. Table 2.3 shows the microstructural effects on mechanical properties of various grades of steel.

The effect of quenching and tempering on fracture toughness of plain carbon steels of varying carbon content is shown in Fig. 2.20. Fracture toughness decreases with increase in the yield strength of steel. Quenched steel (martensite) has the highest yield strength and the lowest fracture toughness compared to the other tempered and normalized steels.

Increasing carbon content increases strength and reduces fracture toughness of steel. Quenching the 0.4% C steel produces martensite, which has high strength but extremely low fracture toughness, below the threshold of 15 MPa (2.2 ksi) for structural applications. Tempering the 0.4% C steel reduces its yield strength but increases its fracture toughness.

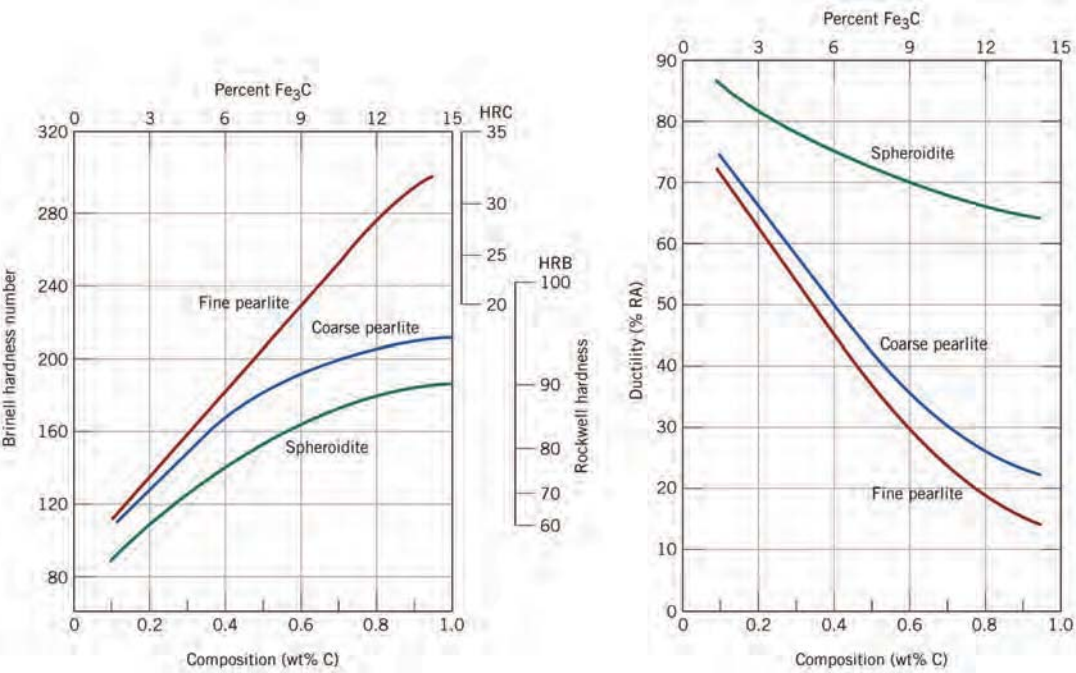


Fig. 2.18 Effect of microstructure on mechanical properties of different plain carbon steel compositions. Source: Ref 2.1

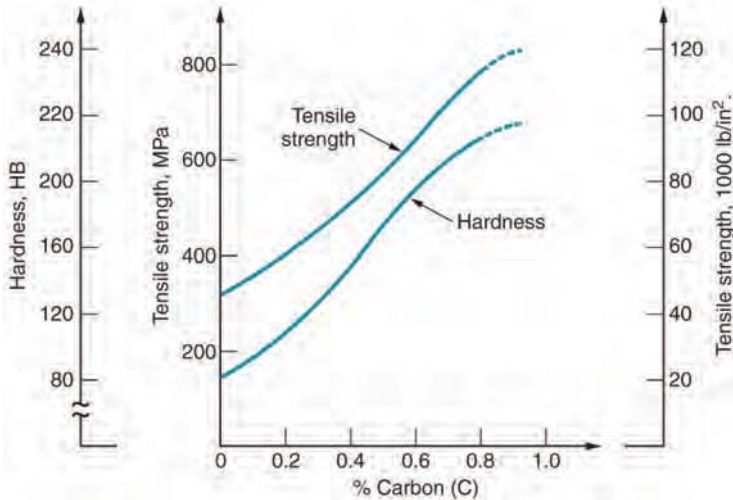


Fig. 2.19 Effect of carbon content on the tensile strength and hardness of carbon steels. Source: Ref 2.1

Table 2.3 Microstructural effects on mechanical properties of various grades of steel

Steel grade	Microstructure	Tensile strength range, MPa (ksi)	Ductility range, %
Interstitial-free (IF)	Ferrite	150–250 (22–36)	40–55
Mild steel	Ferrite + pearlite	180–300 (26–44)	30–45
High-strength, low-alloy (HSLA)	Ferrite + pearlite + alloy precipitates	300–800 (44–116)	10–25
Dual-phase (DP)	Ferrite + martensite	450–1000 (65–145)	6–30
Complex-phase (CP)	Fine ferrite + higher fraction of martensite and pearlite	450–1200 (65–174)	6–30
Transformation-induced plasticity (TRIP)	Soft ferrite + hard bainite + retained austenite	500–1200 (73–174)	10–33
Martensitic (MS)	100% martensite	720–1650 (104–239)	3–12
Twinning-induced plasticity (TWIP)	Retained austenite	1100–1650 (160–239)	43–63
Austenitic stainless steel (AUST SS)	Retained austenite	900–1150 (131–167)	45–63

The higher the tempering temperature, the higher is the fracture toughness and the lower is the yield strength.

Figure 2.21 compares the microstructures of various phases of steel grades. Ferrite and martensite have single-phase microstructures. Dual-phase (DP) steel has a two-phase ferrite and martensite, while transformation-induced plasticity (TRIP) steel has a multiphase ferrite, bainite/martensite, and retained austenite microstructure.

2.4 Categories of Steels

Within a steel category, a distinction should be made as to whether or not a significant amount of alloying elements, other than carbon, is used. A composition of 5% alloying elements is an arbitrary boundary between low-alloy and high-alloy steels. Various elements are added to steels to

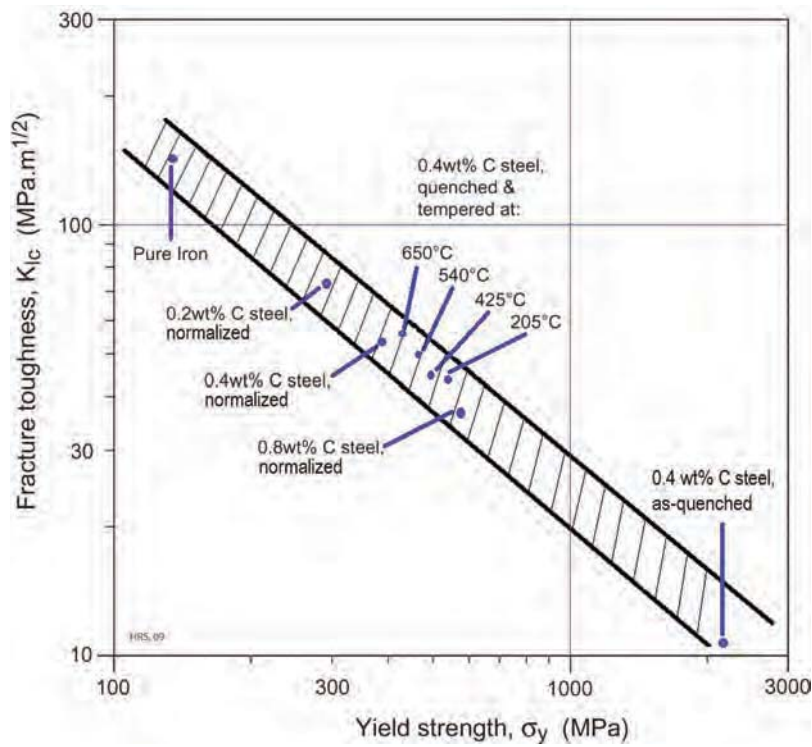


Fig. 2.20 Effect of quenching and tempering on the fracture toughness of plain carbon steels. Source: Adapted from Ref 2.2

produce beneficial properties in strength, hardness, toughness, wear resistance, formability, weldability, and machinability. Alloy additions increase material costs and are justified only by essential improvements in properties for certain applications.

A number of arbitrary definitions of the classes of steels have been developed. The American Iron and Steel Institute (AISI) and the International Ultra-Light Steel Auto Body (ULSAB) Consortium define the strength levels of mild steel, HSS, and AHSS (Table 2.4).

It is important to realize that some HSS may be designated by minimum tensile strength and not by minimum yield strength. To provide a consistent nomenclature, the ULSAB Consortium adopted a standard practice that defines both yield strength (YS) and ultimate tensile strength (UTS).

Original equipment manufacturers (OEMs) use a different categorization scheme for flat rolled steels that is based on the tensile strength of the steel (Table 2.5).

The AISI/SAE and UNS designation systems and composition ranges for plain carbon and low alloy steels are listed in Table 2.6.

Following is a brief description of relevant steel grades (Ref 2.5).

Carbon steels are the most widely used metals and have a wide range of applications. They are characterized by good strength, ductility, weld-

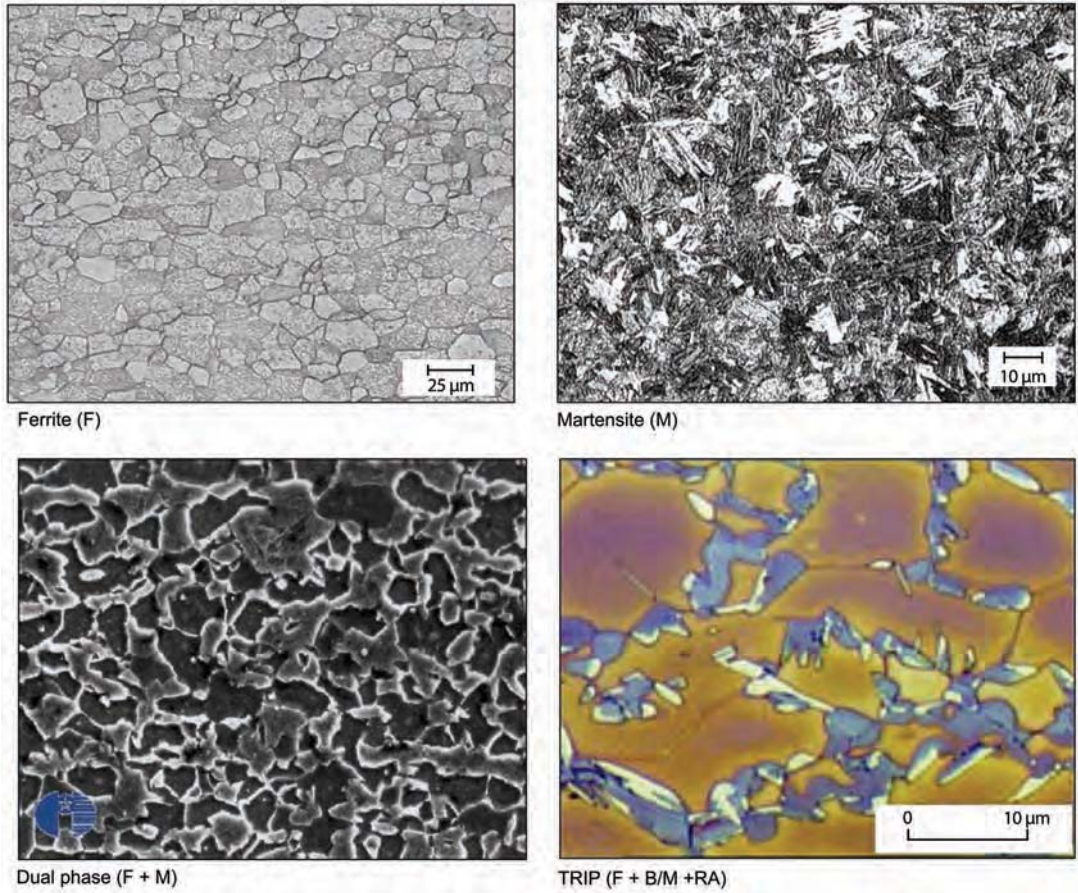


Fig. 2.21 Microstructures of various phases of steel. Source: Ref 2.4

Table 2.4 AISI and ULSAB definitions of mild, high-strength, and advanced high-strength steels

Steel category	Yield strength, MPa (ksi)
Mild steel	140–<210 (20–<30)
High-strength steel (HSS)	210–550 (30–80)
Advanced high-strength steel (AHSS)	>550 (>80)

AISI, American Iron and Steel Institute. ULSAB, International Ultra-Light Steel Auto Body Consortium. Source: Ref 2.5

Table 2.5 Segmentation of flat rolled steel grades by original equipment manufacturers

Steel category	Tensile strength, MPa (ksi)
Mild steel	<270 (39)
Conventional high-strength steel (HSS)	350–590 (51–86)
Advanced high-strength steel (AHSS)	>700 (102)

Table 2.6 Designation systems and composition ranges for plain carbon and low-alloy steels

AISI/SAE designation(a)	UNS designation	Composition ranges (wt% of alloying elements in addition to C)(b)			
		Ni	Cr	Mo	Other
10xx, plain carbon	G10xx0				
11xx, free machining	G11xx0				0.08–0.33 S
12xx, free machining	G12xx0				0.10–0.35 S, 0.04–0.12 P
13xx	G13xx0				1.60–1.90 Mn
40xx	G40xx0			0.20–0.30	
41xx	G41xx0		0.80–1.10	0.15–0.25	
43xx	G43xx0	1.65–2.00	0.40–0.90	0.20–0.30	
46xx	G46xx0	0.70–2.00		0.15–0.30	
48xx	G48xx0	3.25–3.75		0.20–0.30	
51xx	G51xx0		0.70–1.10		
61xx	G61xx0		0.50–1.10		0.10–0.15 V
86xx	G86xx0	0.40–0.70	0.40–0.60	0.15–0.25	
92xx	G92xx0				1.80–2.20 Si

(a) The carbon concentration, in weight percent $\times 100$, is inserted in the place of “xx” for each specified steel. (b) Except for 13xx alloys, Mn concentration is <1.00 wt%. Except for 12xx alloys, P concentration is <0.35 wt%. Except for 11xx and 12xx alloys, S concentration is <0.04 wt%. Except for 92xx alloys, Si concentration varies between 0.15–0.35 wt%. Source: Ref 2.1

ability, and low cost. Carbon steels are classified according to their carbon content. Low-carbon steel has $<0.3\%$ C, medium-carbon steel has 0.30 to 0.6% C, and high-carbon steel has $>0.6\%$ C. The higher the carbon content of the steel, the higher are its strength and hardness and the lower its ductility, toughness, and weldability.

Alloy steels are basically carbon steels with certain chemical elements added to improve the properties of the metal for specific applications. Alloying elements include manganese, silicon, copper, sulfur, phosphorus, nickel, selenium, molybdenum, vanadium, and chromium in amounts totaling less than $\sim 5\%$ by weight. Their mechanical properties, such as strength, hardness, creep resistance, and fatigue resistance are superior to plain carbon steels. One of the reasons for adding alloying elements is to enhance the metal hardenability. Heat treatments, such as quenching and tempering, normalizing, annealing, and stress relieving, are often required to achieve these improved properties. Alloy steels are mainly used in the construction and transportation industries.

Bake-hardenable (BH) steels have a good combination of strength and formability. This combination makes them ideal for dent-resistant auto applications such as hoods, doors, and fenders. The increase in yield strength in this class of steel results from work hardening and strain aging during paint-baking after the stamping process. Bake-hardening produces an increase in yield strength ranging from 28 to 56 MPa (4 to 8 ksi) after a low-temperature heat treatment cycle similar to a paint-bake schedule (175°C , or 350°F , for 20 to 30 min). The bake-hardening effect is possible only if the steel has undergone low plastic deformation of $<5\%$ and has been exposed to a baking temperature of at least 150°C (300°F). These steels are appropriate for some low-deformation body panels.

High-strength, low-alloy (HSLA) steels have low carbon content ($<0.3\%$ C) and approximately 1% or less of alloying elements such as manganese, phosphorus, chromium, nickel, silicon, or molybdenum. Their

microstructure consists of a fine-grain ferrite phase and a hard martensite and austenite phase. The high strength of the HSLA steels results from carefully controlled microalloying and thermal processing. These steels are used mainly in the automotive industry where strength-to-weight ratio applications are important.

High-strength, low-alloy steels have small amounts of alloying elements such as phosphorus, manganese, or silicon added to low-carbon (0.02 to 0.13% C) steels to attain high strength levels. In these steels, higher strength is achieved by rapid cooling to produce a very fine ferrite grain size, by solid-solution strengthening with phosphorus, nitrogen, silicon, and manganese and by formation of carbides or carbonitrides with vanadium, nickel, and titanium. These steels have better mechanical properties, corrosion resistance, and weldability than mild steels. High-strength, low-alloy steels are produced as hot rolled products and cold rolled sheet. These steels can achieve yield strengths up to 485 MPa (70 ksi) without heat treatment. This class of steel has low formability and is used for shallow drawn parts.

Stainless steels are iron-base alloys containing a minimum of 12 to 30% Cr and >50% Fe. Carbon is usually present in amounts ranging from <0.03% to >1.0% in certain martensitic grades. Other alloying elements such as nickel, molybdenum, copper, titanium, manganese, cobalt, aluminum, nitrogen, and sulfur are added to enhance specific characteristics. Stainless steels are characterized by their corrosion resistance, high strength, and ductility. Chromium forms a thin impermeable oxide film that protects the surface from corrosion. Nickel is another alloying ingredient in certain stainless steels that increases corrosion protection. The amount of nickel determines whether the stainless steel is heat treated or remains austenitic. Carbon is used to strengthen and harden stainless steels, but high carbon content reduces corrosion protection because chromium carbide forms and this reduces the availability of the corrosion-resisting free chromium. Stainless steels types and compositions are classified according to the predominant phase present at ambient temperature. They are commonly divided into five groups:

Martensitic stainless (MS) steels are alloys of iron, chromium, and carbon. They contain as much as 18% Cr but no nickel. Carbon content may exceed 1%. They are magnetic and are heat treatable. Heat treatment produces martensite that is hard and strong. They are mainly used where strength, hardness, and wear resistance are required.

Ferritic stainless steels are iron alloys containing approximately 15 to 20% Cr, low carbon, and no nickel. They have a bcc crystal structure and are not heat treatable. They are used for decorative trim, sinks, and automotive exhaust systems.

Austenitic stainless steels (AUST SS) are iron alloys containing between 16 to 26% Cr and approximately 8% Ni. They are nonmagnetic and have a fcc crystal structure. Austenite is retained because of the large amounts of austenitizing elements used, such as nickel, manganese, and nitrogen.

They cannot be hardened by heat treatment and are hardened only by cold working. The austenite phase is tough and ductile. Because of their high chromium and nickel content, they are the most corrosion resistant of the stainless group. Also, they have excellent mechanical properties. Typical applications include flatware and sinks, pressure and cryogenic vessels, and food and chemical processing equipment.

Duplex (ferritic-austenitic) stainless steels are alloys of iron containing roughly equal amounts of bcc ferrite and fcc austenite. The percentage of each phase depends on the composition and heat treatment. The primary alloying elements are chromium and nickel. Duplex stainless steels have greater tensile and yield strengths, but lower toughness than austenitic stainless steels. They are more resistant to stress-corrosion cracking than austenitic alloys, yet tougher than ferritic alloys.

Precipitation-hardening stainless steels contain chromium and nickel and may be either austenitic or martensitic in the annealed condition. They develop high strength during heat treatment by utilizing precipitation hardening rather than phase transformation. Originally developed for aerospace applications, they are used in many applications.

Table 2.7 compares the major relative properties of four grades of stainless steel. Austenitic stainless steels have the best formability, corrosion resistance, and weldability compared to the other grades.

Stainless steels are significantly more expensive than plain carbon or low-alloy steels because they contain large amounts of expensive alloying metals. They have been used for many industrial, architectural, chemical, and consumer applications. Austenitic stainless steels have excellent formability with good strength and this makes them potential material for automotive application. Austenitic stainless steels are covered in more detail in Chapter 10 in this book, “Austenitic Stainless Steels (AUST SS).”

Advanced high-strength steels are multiphase steels that contain a ferrite matrix and islands of martensite, bainite, austenite, and/or retained austenite in quantities and combinations sufficient to produce desired mechanical properties. They are produced by nontraditional thermal cycles and exhibit a superior combination of high strength with good formability and high strain-hardening capacity. Advanced high-strength steels fall in

Table 2.7 Comparison of relative properties of various stainless steel grades

Property	Stainless steel grade			
	Ferritic 51400	Martensitic 51400	Austenitic 30300	Precipitation hardening
Heat treatable	No	Yes	No	Yes
Magnetic	Yes	Yes	No	Yes
Microstructure	Ferrite	Martensite	Austenite	Martensite
Formability	B	D	A	C
Strength	D	A	C	B
Corrosion resistance	C	D	A	B
Oxidation resistance	A	D	B	D
Weldability	B	D	A	D

A = highest, D = lowest. Source: Ref 2.6